Foraging Behaviour Model Using Reinforcement Learning

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Abstract

This project models animal foraging as a reinforcement learning problem. A simulated animal chooses between collecting food at its current patch and moving to another patch. Food is depleted when collected and regenerates over time, while movement incurs an energy cost. A tabular Q-learning agent is trained and evaluated against random and heuristic strategies.

1. Introduction

Foraging is the process by which an animal searches for and obtains food. Animals balance the benefit of collecting food against the cost of travelling. A food patch may become less rewarding as resources are consumed. Reinforcement learning models this through interaction, actions and reward feedback.

2. Problem Formulation

Element	Description
Agent	A simulated animal selecting foraging actions.
Environment	Two food patches, A and B, with finite capacity and gradual regeneration.
State	(current location, food level in patch A, food level in patch B).
Actions	Forage at the current patch; move to the other patch.
Reward	Food collected; movement incurs an energy penalty of 0.5.
Goal	Maximize cumulative reward over 60 time steps.

3. Environment and Methodology

Patch A has a maximum food level of 10 units and patch B has 8 units. Foraging collects up to 2 units from the current patch. After each action, both patches regenerate by 1 unit, up to their maximum capacities. Moving costs 0.5 reward units. Each episode lasts 60 steps.

The Q-learning agent stores values for each discrete state and action. It uses epsilon-greedy action selection: with probability epsilon it explores a random action; otherwise it selects the action with the highest Q-value. Epsilon decays during training. The update is $Q(s,a) \leftarrow Q(s,a) + \alpha[r + \gamma \max Q(s',a') - Q(s,a)]$. Parameters: learning rate $\alpha=0.15$, discount factor $\gamma=0.95$, initial epsilon=0.20, and 1,200 training episodes.

4. Baseline Strategies

Random baseline: chooses forage or move with equal probability. Heuristic baseline: stays at the current patch unless its food level is 2 or less and the other patch has at least 2 more units. This is an optimal-foraging-inspired rule, not a formal implementation of the Marginal Value Theorem.

5. Experimental Results

The learned policy and baselines were evaluated for 200 episodes of 60 steps each. Results are mean cumulative reward ± standard deviation:

Strategy	Mean reward	Std. deviation
Q-learning	100.00	0.00
Random	45.28	9.19
Heuristic	102.50	0.00

The Q-learning training curve reached a final 40-episode moving average of 97.24. Q-learning outperformed random action selection, while the heuristic achieved a slightly higher evaluation mean. This shows that a learned policy does not automatically outperform a well-matched hand-designed rule in a small deterministic task.

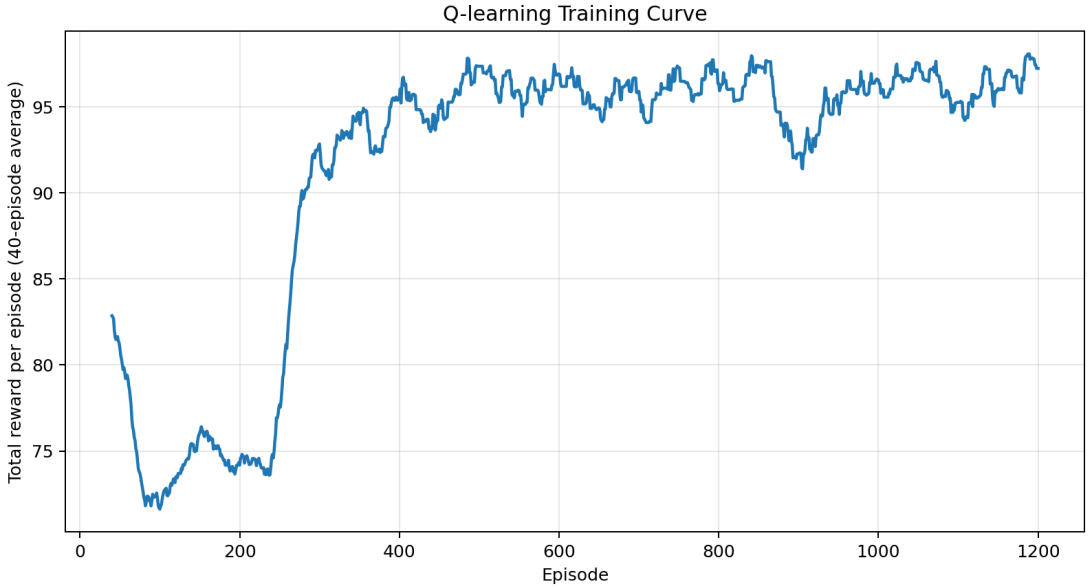


Figure 1. Q-learning training reward (40-episode moving average).

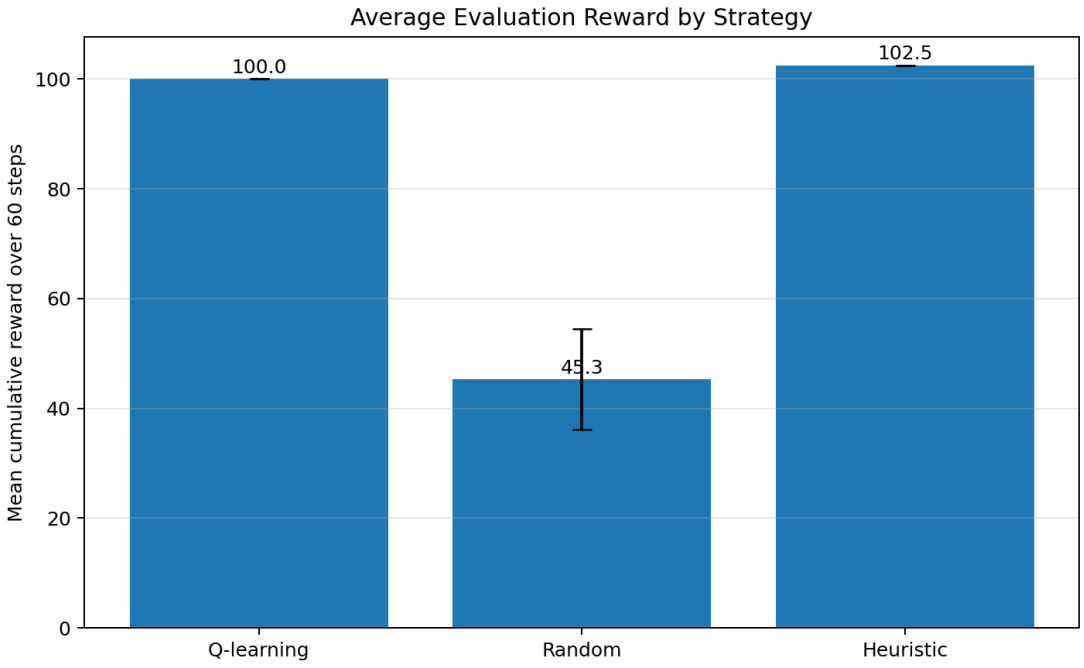


Figure 2. Mean evaluation reward with standard-deviation error bars.

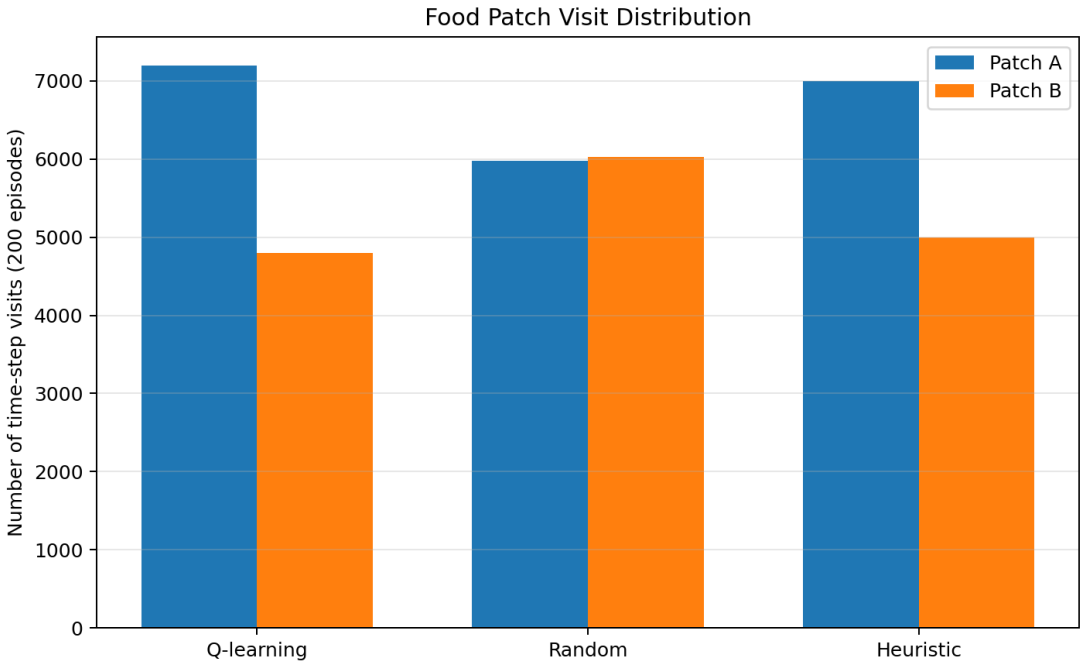


Figure 3. Food patch visit distribution.

6. Discussion

Random actions often cause unnecessary movement and inconsistent food collection. Q-learning improves through repeated interaction and obtains substantially more reward than random behaviour. The heuristic performs best in this setup because its rule is closely matched to the simple food depletion pattern. Results should not be generalized to complex natural environments.

7. Limitations

- Only two food patches and two actions are modelled.
- Food regeneration is deterministic and identical for both patches.
- The agent observes exact food levels; real animals may have incomplete information.
- The heuristic is simple and designed around this environment.
- Predators, weather, competition and biological energy constraints are omitted.

8. Ethical Considerations

This is a computer simulation and does not involve live animals, animal testing or personal data. It is intended for education. Biological interpretations should be cautious because real ecological behaviour is more complex.

9. Conclusion

The project demonstrates a basic RL formulation for foraging. A Q-learning agent learns to choose between collecting food and moving between patches. It performs better than random behaviour, although a hand-designed heuristic scores slightly higher in this simplified environment. Future work could add more patches, uncertain observations, variable regeneration and a formal Marginal Value Theorem benchmark.

References

1. Sutton, R. S., and Barto, A. G., Reinforcement Learning: An Introduction, 2nd ed., MIT Press, 2018.

2. Charnov, E. L., “Optimal foraging, the marginal value theorem,” Theoretical Population Biology, vol. 9, no. 2, pp. 129–136, 1976.

3. Python, NumPy and Matplotlib documentation.

Additional Baseline Learning-Curve Comparison

Figure 4 compares the Q-learning agent with random and heuristic reference policies across training episodes. The plotted values are 40-episode moving averages. The random and heuristic policies are fixed baselines and do not update from experience; their curves provide reference performance levels against which the Q-learning agent's learning progress can be assessed.

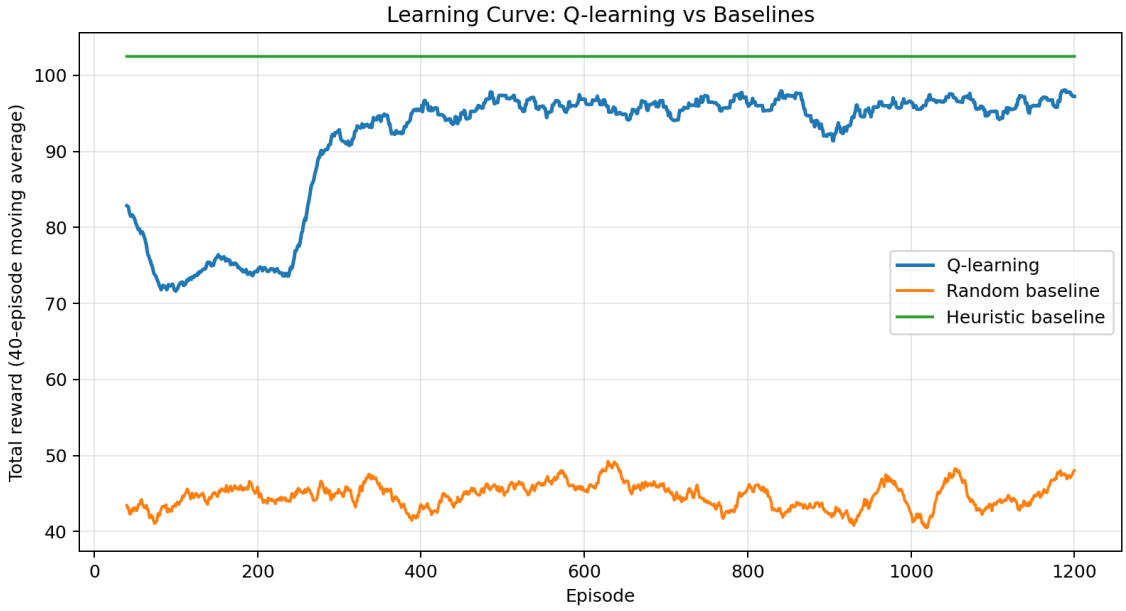


Figure 4. Q-learning learning curve compared with random and heuristic baselines.